

Keshav Bansal

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LinkedIn — GitHub — Portfolio

EDUCATION

Thapar Institute of Engineering and Technology	2023 – Present
B.E. in Computer Engineering	CGPA: 7.63
Rosemary Public School, Punjab	2023
Class XII (CBSE)	77.6%
Rosemary Public School, Punjab	2021
Class X (CBSE)	85.8%

TECHNICAL SKILLS

Languages: C/C++, Python, HTML, CSS, SQL/PLSQL, MATLAB

Tools: Jupyter Notebook, VS Code, Kali Linux, VirtualBox, Ubuntu, EC-Council Labs

CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOPs), Operating Systems (OS), Database Management Systems (DBMS), Network Defense, Computer Networks, Machine Learning

Cybersecurity Tools: Nmap, Wireshark, Burp Suite, Metasploit

PROJECTS

Ransomware Detector GitHub

- Built a hybrid ML–DL ensemble to detect and classify 26 ransomware families.
- Utilized static and dynamic features for ransomware detection.
- Applied model stacking techniques achieving approximately 97% accuracy.

PhishGuard – AI-Driven Phishing Detection

GitHub

- Implemented phishing detection using TF-IDF with Logistic Regression, Naive Bayes and Random Forest.
- Integrated SBERT embeddings for contextual email understanding.
- Designed a recall-oriented hybrid pipeline and generated synthetic phishing data using GPT-2.

AI-Based Attendance System

GitHub

- Developed a real-time facial recognition attendance system using OpenCV and scikit-learn.
- Automated training, testing and user registration workflows.
- Optimized live inference for attendance marking.

Intrusion Detection System (DNS)

GitHub

- Built an end-to-end machine learning pipeline for malicious DNS traffic detection.
- Processed DNS queries using Regex and TF-IDF feature extraction.
- Implemented real-time detection with latency tracking and performance evaluation.

CYBERSECURITY PLATFORMS

- TryHackMe (Beginner–Intermediate Labs)
- EC-Council Cybersecurity Labs
- Kali Linux based testing environments using VirtualBox

CERTIFICATIONS

- Summer Internship: Phishing Detection Using Blockchain – TIET
- Cybersecurity Essentials – Cisco Networking Academy
- Network Defense – Cisco Networking Academy
- Ethical Hacking – Cisco Networking Academy